Negative exponents

1 Express the following using only positive exponents:

a p^{-2} **b** a^{-8} **c** $3x^{-2}$ **d** $10x^{-6}$

e $4a^{-2}$ **f** $4y^{-3}$ **g** $\dfrac{1}{a^{-n}}$ **h** $\dfrac{a^{-n}}{b^{-m}}$

i $p^{-2}q^{3}$ **j** $8p^{-3}$ **k** $m^{-5}n^{-4}p^{4}$

2 Express the following using only negative exponents:

a $\dfrac{1}{u^{5}}$ **b** $\dfrac{10}{p^{8}q^{5}}$

3 Write an expression for the reciprocal of a^{3}.

4 Find the value of n on the following equations:

a $\dfrac{1}{36} = 6^{n}$ **b** $\dfrac{1}{8} = 2^{n}$

5 Determine the missing exponent so that following equation is always true.

$$\left(a^{3}b^{-5}\right)^{\square} = a^{-15}b^{25}$$

6 Simplify the following, expressing your answers using positive exponents:

a $4y^{-5} \cdot 3y^{-3}$ **b** $(6m)^{2} \cdot m^{-6}$

c $2y^{3} \cdot \left(-2y^{-7}\right)$ **d** $6y^{7} \cdot 2y^{-5} \cdot 5y^{3}$

e $5p^{3}q^{-2} \cdot 9p^{-3}q^{5}$ **f** $5p^{7}q^{-7} \cdot 5p^{-7}q^{3}$

g $8p^{5}q^{-5} \cdot 4p^{-4}q^{3}$ **h** $2p^{4}q^{-2} \cdot 5p^{-4}q^{-5}$

i $5p^{-6}q^{-9} \cdot \left(-6p^{-4}q^{-3}\right)$ **j** $\left(w^{4}\right)^{-5}$

k $\left(w^{-9}\right)^{5}$ **l** $\left(5^{2}\right)^{-p}$

m $\left(4y^{4}\right)^{-4}$ **n** $\left(2y^{-5}\right)^{4}$

o $\left(-4u^{-4}\right)^{3}$ **p** $\left(-u^{3}\right)^{-4}$

q $\left(4m^{-10}\right)^{4}$ **r** $\left(4m^{-8}\right)^{-3}$

s $(4m)^{-3}$ **t** $\left(5y^{3}\right)^{-3}$

u $\left(4p^{8}q^{4}\right)^{-2}$ **v** $\left(\dfrac{4}{5}u^{-3}\right)^{4}$

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w $\left(\frac{y}{2}\right)^{-5}$

7 Assuming that all variables are non-zero, simplify the following. Express your answers using positive exponents.

a $\frac{12x^3}{4x^7}$

b $\frac{6x^3}{2x^{-3}}$

c $\frac{25x^{-7}}{5x^{-4}}$

d $\left(\frac{a}{b}\right)^{-5}$

e $\left(\frac{a^3}{b^3}\right)^{-5}$

f $\left(\frac{a^2}{b^5}\right)^{-1}$

g $\left(\frac{3a^9}{b^9}\right)^{-2}$

h $\frac{-4x^2}{2x^{-5}}$

i $\frac{-6x^{-7}}{-2x^{-4}}$

j $\frac{z^{-2}x^{-5}}{y^{-7}}$

k $\frac{\left(6y^3z^2\right)^{-4}}{\left(6y^3z^2\right)^{-5}}$

l $\frac{5p^5q^{-4}}{40p^5q^6}$

m $\left(\frac{2y^3x^0}{z^2}\right)^{-4}$

n $\frac{x^2y^{-3}}{x^{-5}y^2}$

o $\frac{\left(m^{-3}\right)^{-1}\cdot\left(m^4\right)^{-3}}{m^3\cdot m^4}$

p $\left(\frac{m^7}{m^{-10}}\right)^2\cdot\left(\frac{m^5}{m^2}\right)^{-3}$

q $\frac{a^2\cdot a^{-5}\cdot b^{-2}}{\left(a\cdot a^{-5}\cdot b^2\right)^2}$

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